Joint Knowledge Distillation and Tensor-Based Compression of Deep Neural Networks

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1 Introduction

Deep neural networks deliver high accuracy but carry millions of parameters and billions of FLOPs per inference, far exceeding the memory and power budgets of edge platforms such as autonomous drones, IoT sensors and medical devices. This bottleneck is the memory wall [1]; fetching model parameters from DRAM eclipses available compute, making aggressive compression an engineering imperative for practical edge AI.

Two compression families exist yet each fails in isolation. **Tensor decomposition** (CP [2], Tucker [3]) reduces structural size but causes vanishing gradients and accuracy collapse in the sequential decompose-then-fine-tune pipeline. **Knowledge distillation** [5] guides a compact student with a teacher's soft outputs but collapses at high compression ratios due to the capacity gap [6] and becomes structurally ill-defined once the student's layers are tensor-factorized.

2 Problem Definition

Given a pre-trained teacher $\mathcal{T}$ and labelled dataset $\mathcal{D}$, jointly construct and train two compressed students S_{CP}, S_{Tuck} from scratch to minimise the combined distillation loss:

$$\mathcal{L} = \sum_{i \in \{CE, KL, AT\}} \left(e^{-s_i} \mathcal{L}_i + s_i \right)$$

subject to:

- Per-layer ranks $\{r_i\}$ derived by **EVBMF** [10] on $\mathcal{T}$'s weight spectrum — automatic, layer-adaptive, no manual tuning.
- Hint points $\mathcal{H} = \{h_1, \ldots, h_K\}$ selected by **PURSUhInT** [8] clustering of $\mathcal{T}$'s layer representations.
- Log-variance weights s_i learned online (Kendall uncertainty [22]) — zero hand-tuned loss coefficients.

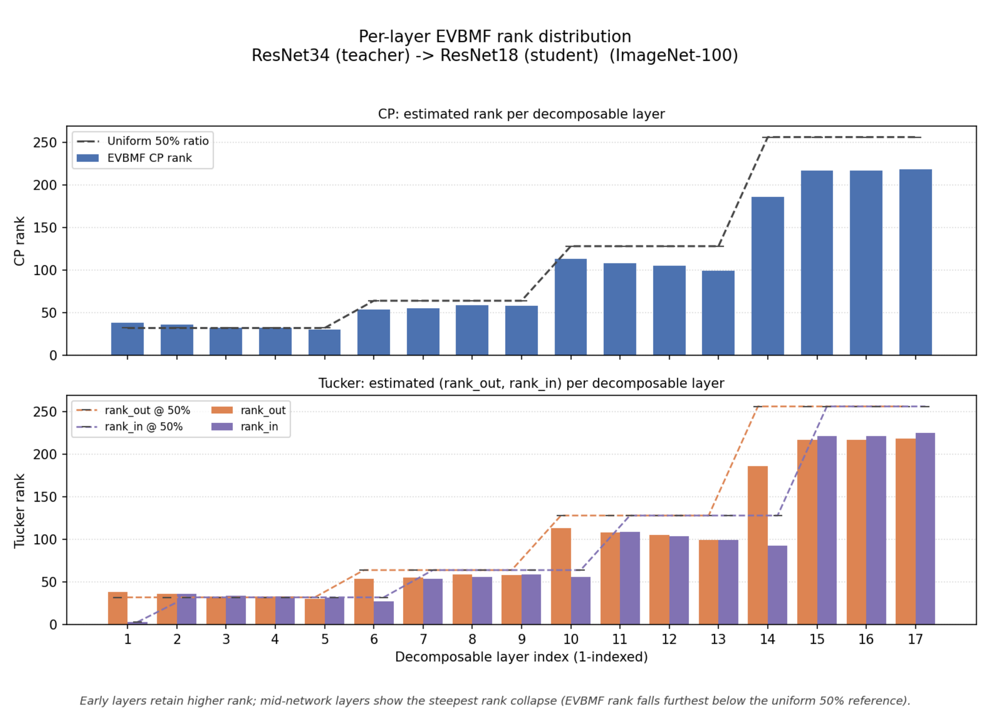
3 Contributions

- **Joint, from-scratch** construction & co-training of CP- and Tucker-decomposed students under one shared teacher, replacing the fragile decompose-then-fine-tune pipeline.
- **PURSUhInT** clustering integrated into compressed-student training to pick non-redundant, semantic hint points.
- **EVBMF** teacher-guided per-layer ranks, automatic, calibrated to the teacher's own weight spectrum.
- **Systematic study**: 8 teacher–student pairs, 2 datasets, 5 distillation objectives, one identical pipeline.

4 Automatic per-layer ranks (EVBMF)

EVBMF [10] reads the teacher's trained weight spectrum to set each layer's rank — no global ratio, no validation search.

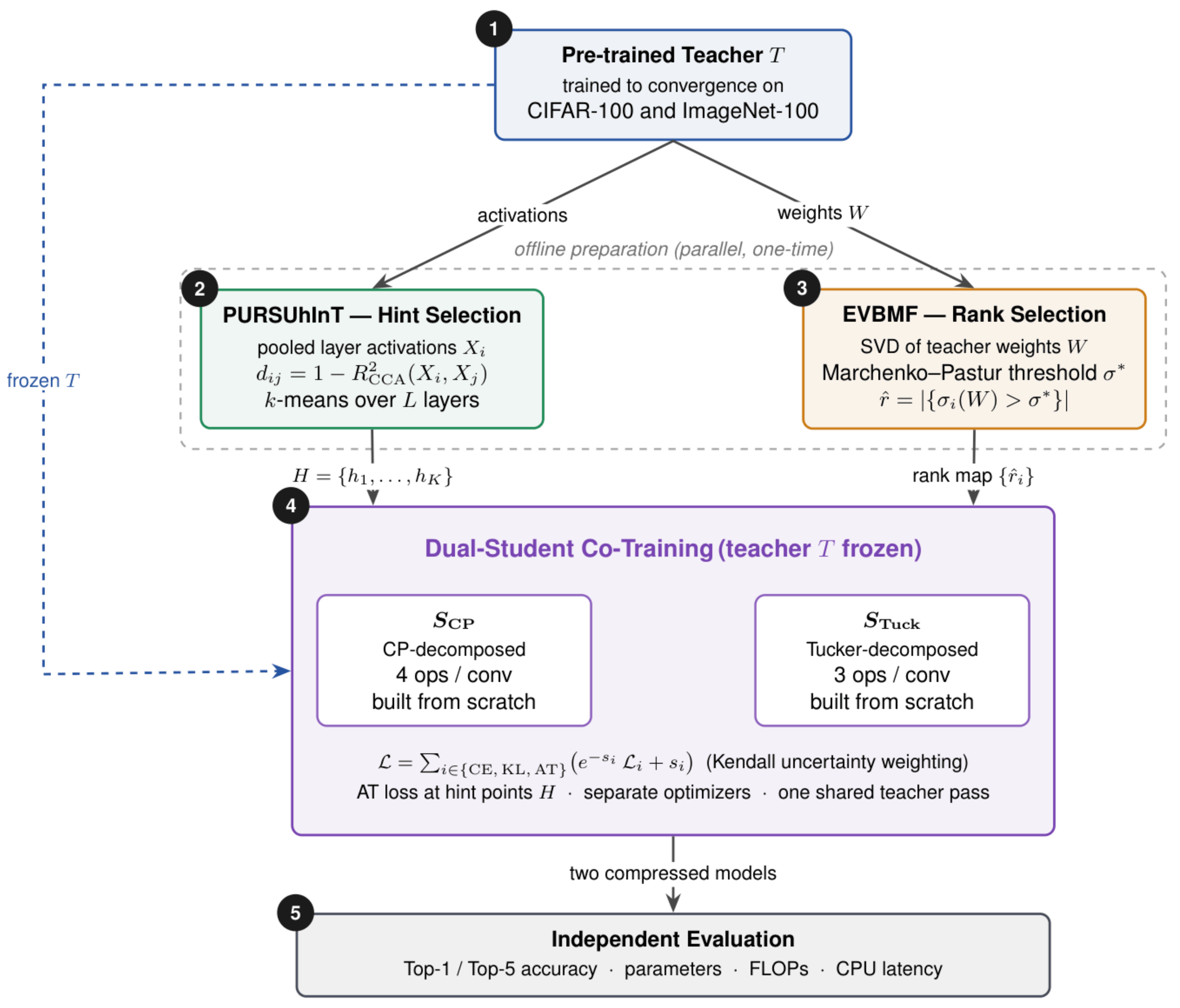
$$\hat{R} = \left| \sigma_i(\mathcal{W}) : \sigma_i(\mathcal{W}) > \sigma^*(\mathcal{W}) \right|$$



Markedly non-uniform: early layers keep $\sim$59% of channels, deep layers fall to 36–44%. Tucker estimates input/output ranks *independently* per stage (Marchenko–Pastur threshold).

5 Proposed Framework

A single joint optimization replaces the sequential pipeline: two compressed students are built from scratch and co-trained under one frozen teacher, then evaluated independently.



One shared teacher forward pass supervises both students each step — roughly twice the cost of a single distillation, not three times.

6 Tensor Decomposition (CP & Tucker)

Every eligible convolution is rebuilt from scratch as a low-rank factorization, with per-layer rank set by EVBMF (block 4). Both students share one base net, factorized two complementary ways:

CP | 4 ops / conv

$$\mathcal{W} \approx \sum_{r=1}^{R} \mathbf{a}_r \otimes \mathbf{b}_r \otimes \mathbf{c}_r \otimes \mathbf{d}_r$$

$$1{\times}1 \to \mathrm{dw}\ k_H \to \mathrm{dw}\ k_W \to 1{\times}1$$

Params $R(C_{in}{+}k_H{+}k_W{+}C_{out})$. Separable spatial filtering ⇒ **highest compression**.

Tucker | 3 ops / conv

$$\mathcal{W} \approx \mathcal{G} \times_1 U_{out} \times_2 U_{in}$$

$$1{\times}1 \to k_H{\times}k_W \text{ core} \to 1{\times}1$$

Params $R_{in}C_{in}{+}R_{in}R_{out}k_Hk_W{+}R_{out}C_{out}$. Dense core keeps cross-channel correlations ⇒ **higher acc. & lower latency**.

7 How it works

PURSUhInT | where to supervise

Cluster teacher layers by representational similarity; each cluster centroid is a non-redundant hint point ($K{=}3$ CIFAR, $K{=}4$ ImageNet).

$$d(i,j) = 1 - R^2_{\mathrm{CCA}}(X_i, X_j) \xrightarrow{k\text{-means}} \mathcal{H} = \{h_1, \ldots, h_K\}$$

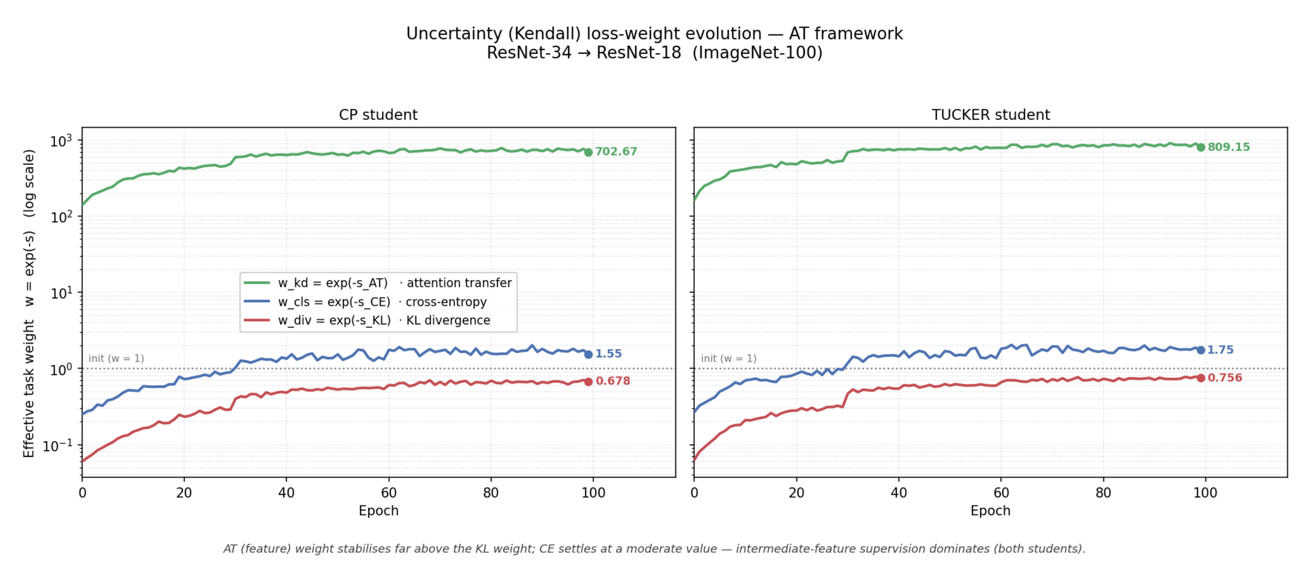
Distillation + auto-balancing | *how* to train

KD + attention transfer at the hint points:

$$\mathcal{L}_{\mathrm{KL}} = \mathrm{KL}\big(\sigma(z_t/\tau) \,\|\, \sigma(z_s/\tau)\big), \quad \mathcal{L}_{\mathrm{AT}} = \sum_k \mathrm{MSE}(a_s^k, a_t^k)$$

Homoscedastic (Kendall) uncertainty weighting — no hand-tuned coefficients:

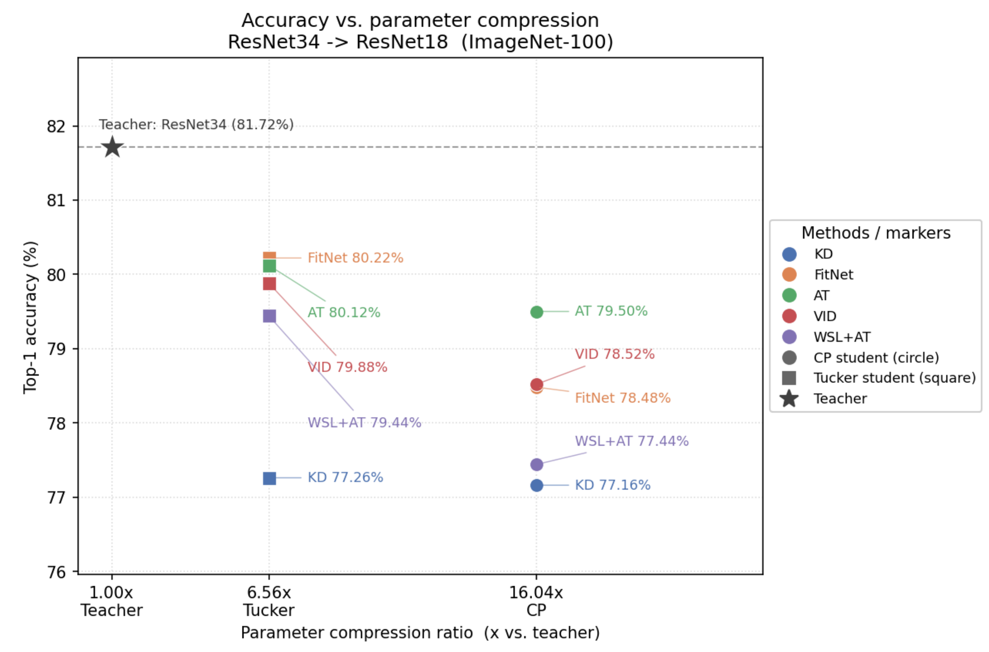
$$\mathcal{L} = \sum_{i \in \{\mathrm{CE, KL, AT}\}} \left(e^{-s_i} \mathcal{L}_i + s_i \right)$$



Weights span 3 orders of magnitude yet equalize the three gradient contributions to within 10% (CE 0.90, KL 1.00, AT 0.91).

8 Key Results

8 teacher–student pairs on CIFAR-100 & ImageNet-100, each under 5 distillation objectives — KD [5], FitNet [7], AT [11], VID [19], WSL [18] in one identical pipeline.



Teacher → Student	Dec.	Compr.	Top-1 (%) Tchr.	Top-1 (%) Stud.
ImageNet-100				
ResNet-34 → 18	Tucker	6.56×	81.72	80.12
	CP	16.04×		79.50
ResNet-50 → 18	Tucker	7.46×	82.54	**81.04**
	CP	17.59×		79.60
CIFAR-100				
WRN-40-2 → WRN-16-2	Tucker	6.78×	76.79	72.23
	CP	20.27×		71.85

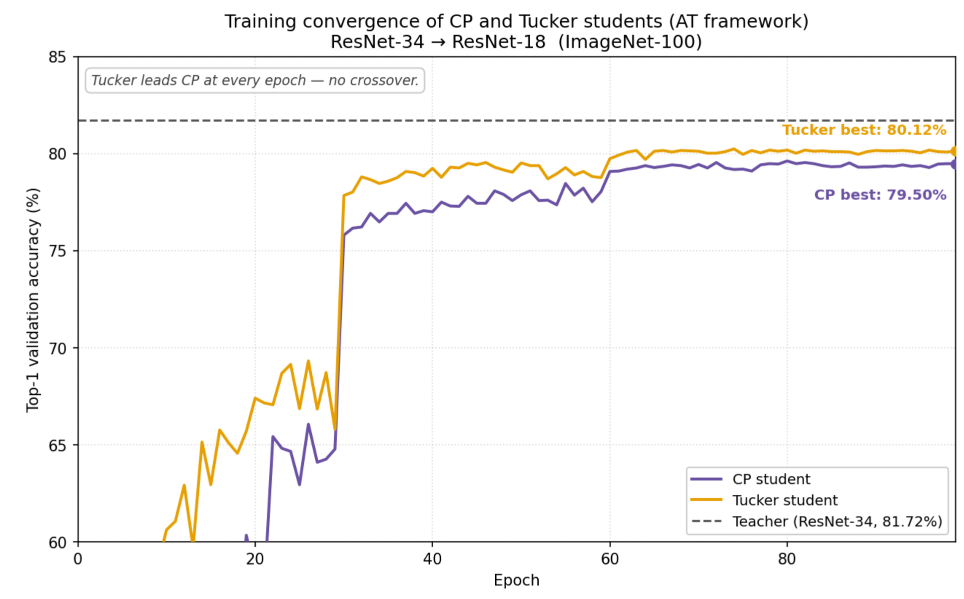
Top-1 shown is under **attention transfer**, the framework's default objective (best in **10 of 16** columns)

9 Efficiency & CP–Tucker trade-off

The two students form a consistent trade-off and the ordering never reverses.

	CP	Tucker
Compression	up to **64**×	up to **22**×
FLOPs compr.	up to **52**×	up to **15**×
Accuracy	lower	higher
CPU latency	1.96×	**2.11**×

Single-image CPU latency 265.98 → 135.88 ms (CP) → 125.91 ms (Tucker). Tucker is faster *despite* 2.4× more params — a 3-op vs 4-op kernel chain.



10 Conclusions

- Joint compression + distillation is a **robust alternative to the fragile sequential pipeline** — Tucker stays within **1.6%** of the teacher on ImageNet-100; both students within **3–5%** on moderate-compression CIFAR-100 pairs (**7–8%** at the hardest 64×).
- **Feature-based supervision is essential** under heavy compression: attention transfer is best in **10 of 16** columns and beats logit-only KD in nearly every configuration.
- **Fully automatic** — EVBMF ranks and Kendall loss weighting need **no hand-tuned rank ratios or loss coefficients**.
- **Tucker** for accuracy & lowest latency (faster *despite* 2.4× more params); **CP** for the tightest budget.

Future Work: vision transformers · full ImageNet-1k · quantization & pruning · on-device energy profiling.

References

Please see the paper via the QR Code for the references as well as for details of notations, other experiments and further explanation.

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